

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I S MD Abdur Rahman Ghouse(192311197) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *S MD Abdur Rahman Ghouse*

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

CONCEPT MAPPING SOLUTIONS

1. Missing OSI layers:

Completed concept map:

Application → Presentation → Session → Transport → Network → Data Link → Physical

The missing layers are Session Layer and Data Link Layer.

Session Layer: manages sessions, synchronization, and recovery during communication.

Data Link Layer: provides framing, MAC addressing, error detection, and local delivery between directly connected devices.

These layers are important because Session maintains continuous communication, while Data Link ensures reliable node-to-node transfer.

2. Data units with OSI layers

Concept map:

Application/Presentation/Session → Data

Transport Layer → Segment

Network Layer → Packet

Data Link Layer → Frame

Physical Layer → Bits

Data is transformed step by step. The Transport Layer divides data into segments, the Network Layer adds IP addressing to form packets, the Data Link Layer adds MAC address and creates frames, and the Physical Layer converts frames into bits for transmission.

3. MAC address and IP address delivery

Concept map:

Data Link Layer → MAC Address → Local Delivery

Network Layer → IP Address → Global Delivery

The Data Link Layer uses MAC addresses to deliver frames within the same local network. The Network Layer uses IP addresses to route packets across different networks. Together, MAC addressing handles local delivery, while IP addressing ensures end-to-end delivery across the internet.

4. Error control in Data Link and Transport Layers

Concept map:

Error Control → Data Link Layer → Error Detection

Error Control → Transport Layer → Error Correction

The Data Link Layer detects errors in frames using methods like CRC. The Transport Layer provides correction through acknowledgements, retransmission, sequencing, and flow control. Their interaction improves reliability because damaged data is detected at the link level and lost or incorrect data can be recovered by the transport layer.

5. Web browsing concept map using OSI layers

Concept map:

User enters URL

→ **Application Layer: HTTP/HTTPS request**

→ **Presentation Layer: encryption/decryption using SSL/TLS**

→ **Session Layer: manages connection session**

→ **Transport Layer: TCP segmentation and reliability**

→ **Network Layer: IP addressing and routing**

→ **Data Link Layer: framing and MAC addressing**

→ **Physical Layer: transmission as electrical/wireless signals**

If one layer fails, the whole communication process is affected. For example, if the Network Layer fails, packets cannot be routed, so TCP and the web application will not work even if the lower layers are functioning.

6. Troubleshooting using concept map

Given map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

The Physical and Data Link layers are working, so cables, signals, framing, and MAC-level communication are correct. The problem starts at the Network Layer, which means there may be an IP address, routing, gateway, or DNS-related issue.

Because the Network Layer fails, the Transport Layer cannot establish proper TCP/UDP communication, and the Application Layer cannot access services like websites. This concept map helps troubleshooting by showing that checking should begin from the Network Layer, not from Physical or Data Link layers.